# Solution Architecture

*Project: Automated Network Request Management (ServiceNow)*

## 1. Architecture Overview

This solution is built entirely on the ServiceNow platform using a scoped application, following a data-driven workflow architecture: a Service Catalog front-end feeds structured network access/change requests into standard Requested Item records, which drive an automated Approval and Fulfillment process via Flow Designer and Business Rules.

## 2. Layered Architecture

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│ Presentation Layer                            │
│ - Service Catalog Item (employee-facing form) │
│ - My Approvals (manager-facing)               │
│ - My Requests (employee self-service)         │
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│ Application / Process Layer                   │
│ - Flow Designer (Approval routing, branching) │
│ - Business Rules (Incident auto-creation)     │
│ - Approvals Engine (sysapproval_approver)     │
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│ Data Layer                                    │
│ - sc_cat_item / sc_request / sc_req_item      │
│ - sc_task (fulfillment)                       │
│ - incident (capacity/security escalation)     │
│ - sys_user (employee / manager references)    │
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```

## 3. Component Interaction Diagram

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Employee --(submits)--> Catalog Item

    Catalog Item --(creates)--> Request --> Requested Item

        Requested Item --(triggers)--> Approval Flow

            Approval Flow --(assigns)--> Manager (My Approvals)

                Manager --(decision)--> Approved / Rejected

                    Approved  --> Catalog Task --> Network/IT Support Team
```

```
                    Rejected  --> Email Notification --> Employee

                On Hold (capacity/security conflict) --> Business Rule --> Incident
--> Network Support Group
```

## 4. Key Design Decisions

- Built as a scoped application to isolate all custom logic and enable clean packaging/version control.
- Used native Service Catalog + Approvals engine instead of custom tables, to leverage existing, well-tested ServiceNow infrastructure.
- Flow Designer (low-code) used for orchestration instead of classic Workflow Editor, for maintainability.
- Business Rules used only for simple, deterministic logic (e.g., Incident creation on a state change) — kept outside the main Flow to separate concerns.

## 5. Security Architecture

- Role-based access: Employee, Manager, and Network/IT Support roles define who can view/act on records.
- Approvals are restricted to the specific manager assigned via the Approval Field, preventing unauthorized approval actions.
- Cross-scope access explicitly granted only where required (e.g., email/notification creation from the scoped app).

## 6. Scalability & Extensibility

- Additional approval levels (e.g., network security team review) can be added as further branches in the Flow without redesigning the data model.
- A Network Capacity/Segment Inventory table can be introduced later to replace the simple On Hold/Incident escalation with real-time availability checks.
- The same architecture pattern can be reused for other request types (software, hardware, facilities access, etc.) by creating new Catalog Items pointing to the same Flow pattern.